FET BIASING

DC analysis of JFET, MOSFET and MESFET networks

FET BIASING

The general relationships that can be applied to the dc analysis of all FET amplifiers are

$$I_G \cong 0\ \text{A} \tag{7.1}$$

and

$$I_D = I_S \tag{7.2}$$

For JFETs and depletion-type MOSFETs and MESFETs, Shockley's equation is applied to relate the input and output quantities:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \tag{7.3}$$

For enhancement-type MOSFETs and MESFETs, the following equation is applicable:

$$I_D = k(V_{GS} - V_T)^2 \tag{7.4}$$

FET BIASING

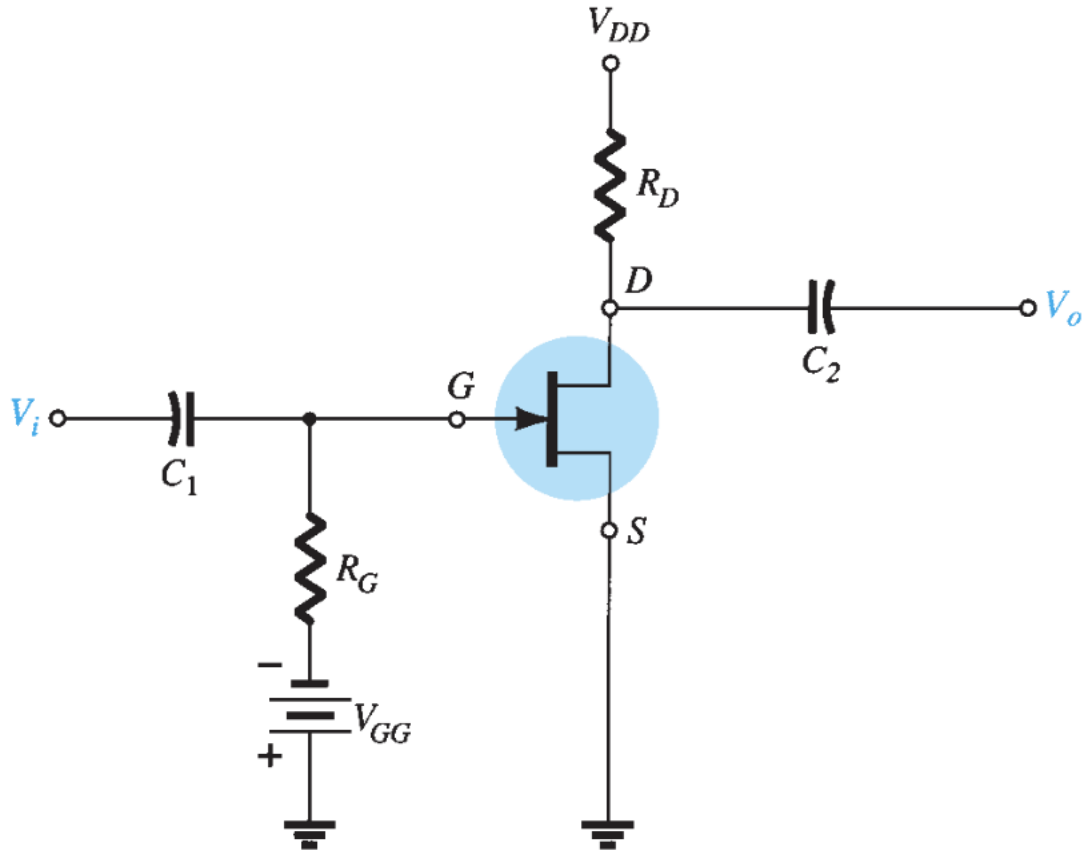


FIG. 7.1
Fixed-bias configuration.

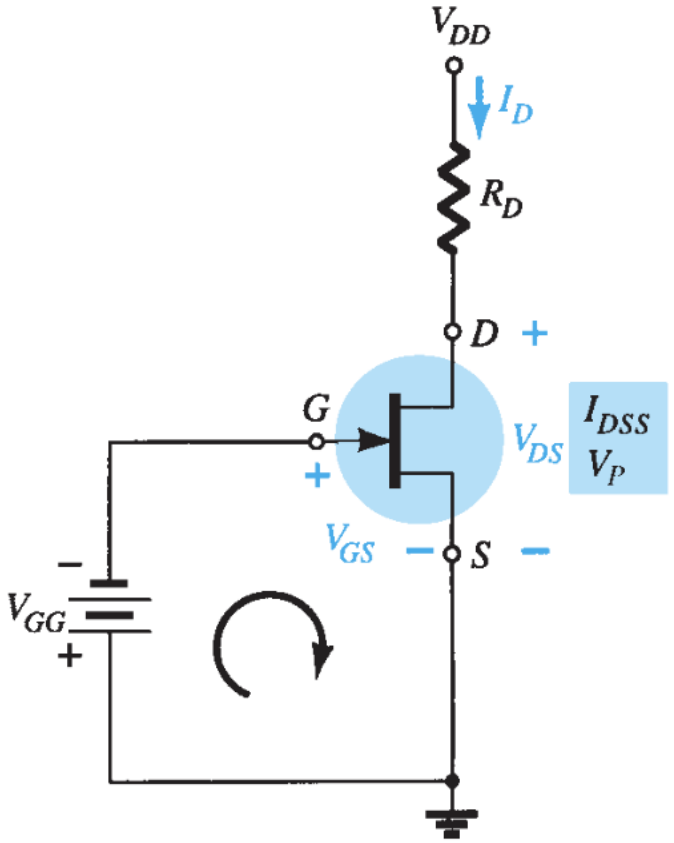


FIG. 7.2
Network for dc analysis.

EXAMPLE 7.1 Determine the following for the network of Fig. 7.6:

a. V_{GS_Q}.
b. I_{D_Q}.
c. V_{DS}.
d. V_D.
e. V_G.
f. V_S.

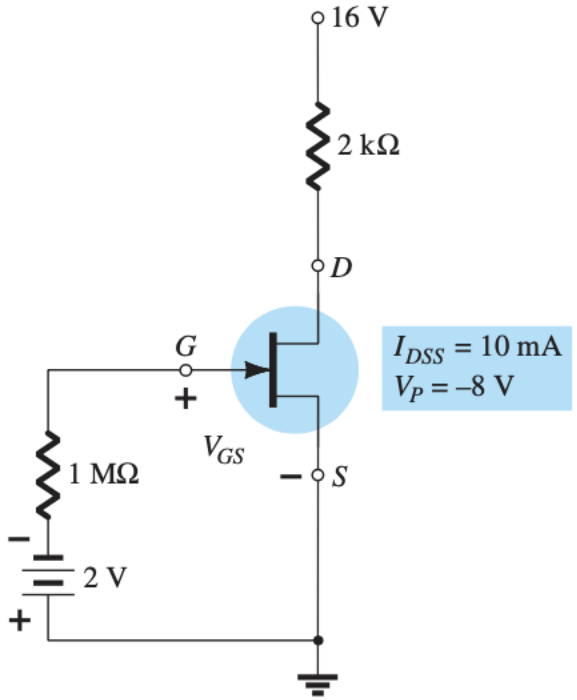


FIG. 7.6
Example 7.1.

a. $V_{GS_Q} = -V_{GG} = \mathbf{-2\ V}$

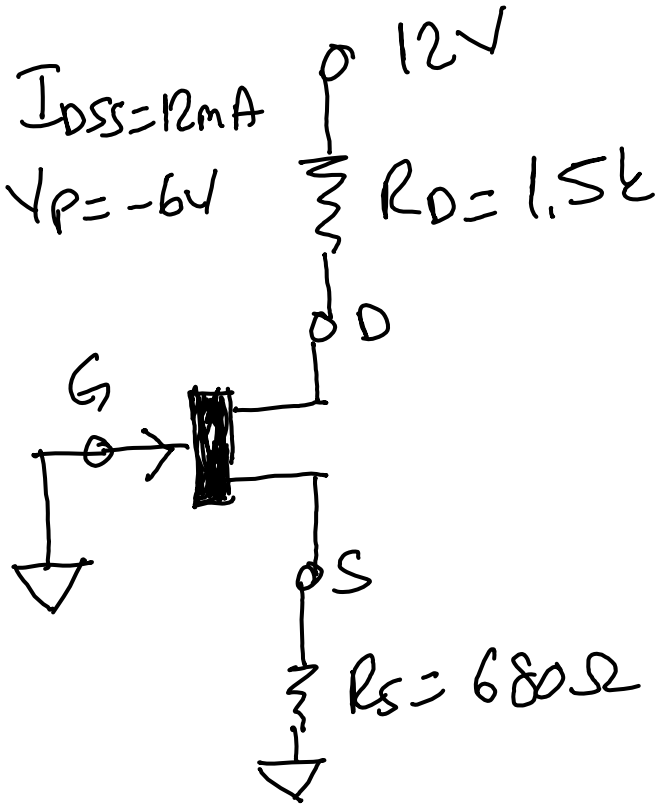
b. $I_{D_Q} = I_{DSS}\left(1 - \dfrac{V_{GS}}{V_P}\right)^2 = 10\text{ mA}\left(1 - \dfrac{-2\text{ V}}{-8\text{ V}}\right)^2$
$= 10\text{ mA}(1 - 0.25)^2 = 10\text{ mA}(0.75)^2 = 10\text{ mA}(0.5625)$
$= \mathbf{5.625\ mA}$

c. $V_{DS} = V_{DD} - I_D R_D = 16\text{ V} - (5.625\text{ mA})(2\text{ k}\Omega)$
$= 16\text{ V} - 11.25\text{ V} = \mathbf{4.75\ V}$

d. $V_D = V_{DS} = \mathbf{4.75\ V}$

e. $V_G = V_{GS} = \mathbf{-2\ V}$

f. $V_S = \mathbf{0\ V}$

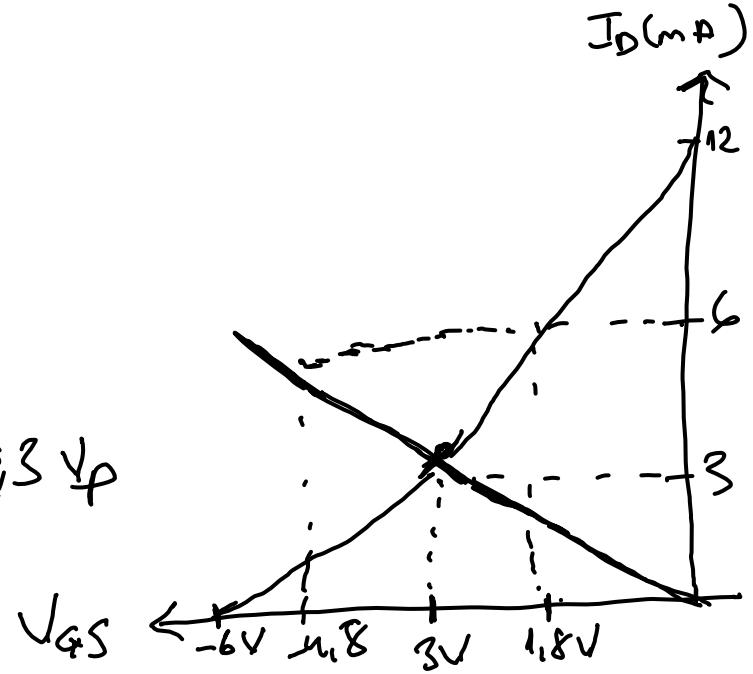


$$I_D = \frac{I_{DSS}}{4}\Big|_{V_{GS}=-3}$$

$$= 3\,mA$$

$$I_D = \frac{I_{DSS}}{2}\Big|_{V_{GS}=0,3\,V_P}$$

$$= 6mA$$



Find $V_{GSQ}, I_{DQ}, V_D, V_G, V_S, V_{DS}$

$V_D = V_{DD} - I_D R_D = 12V - (3,8mA)(1,5K) = 6,3V$

$V_G = 0$

$V_S = I_D . R_S = (3,8mA)(680\Omega) = 2,58V$

$V_{DS} = V_D - V_S = 6,3 - 2,58 = 3,72V$

$V_{GS} = -I_D R_S \Rightarrow$ for $I_D = 6$

$V_{GS} = -4,08$

$V_{GSQ} \sim -2,6V$

$I_{DQ} \sim 3,8mA$

Depletion type

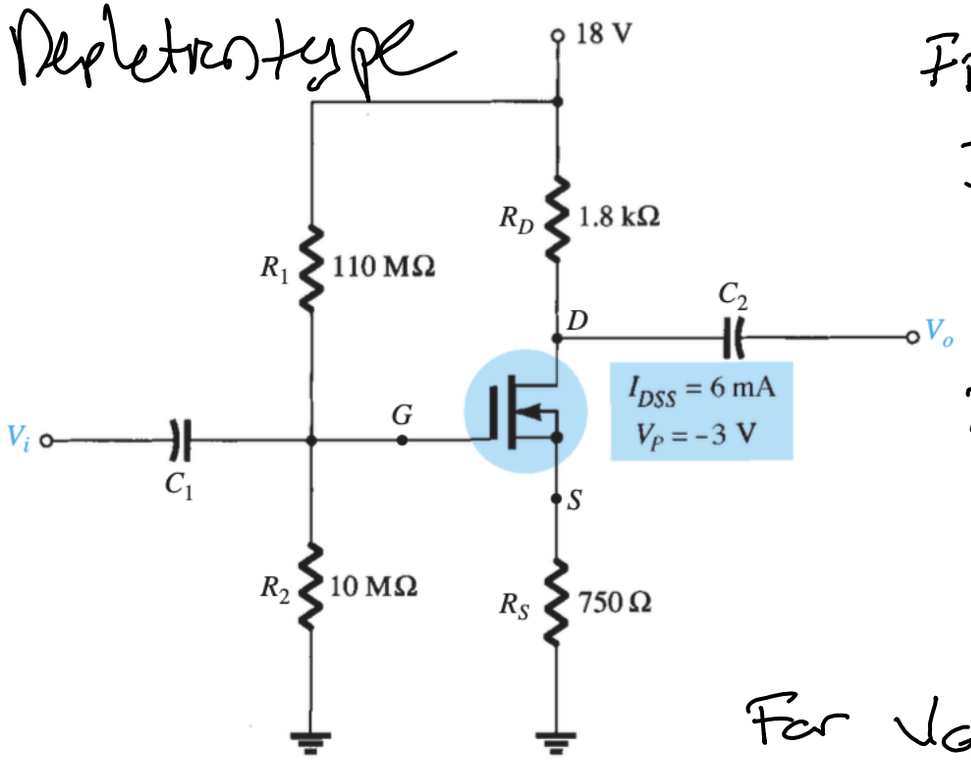


FIG. 7.30
Example 7.6.

First plot the curve

$I_D = \frac{I_{DSS}}{4} = 1,5\,mA$

for $V_{GS} = \frac{V_P}{2} = -1,5V$

$I_D = \frac{I_{DSS}}{2} = 3\,mA$

for $V_{GS} = 0,3\,V_P$

$= -0,4V$

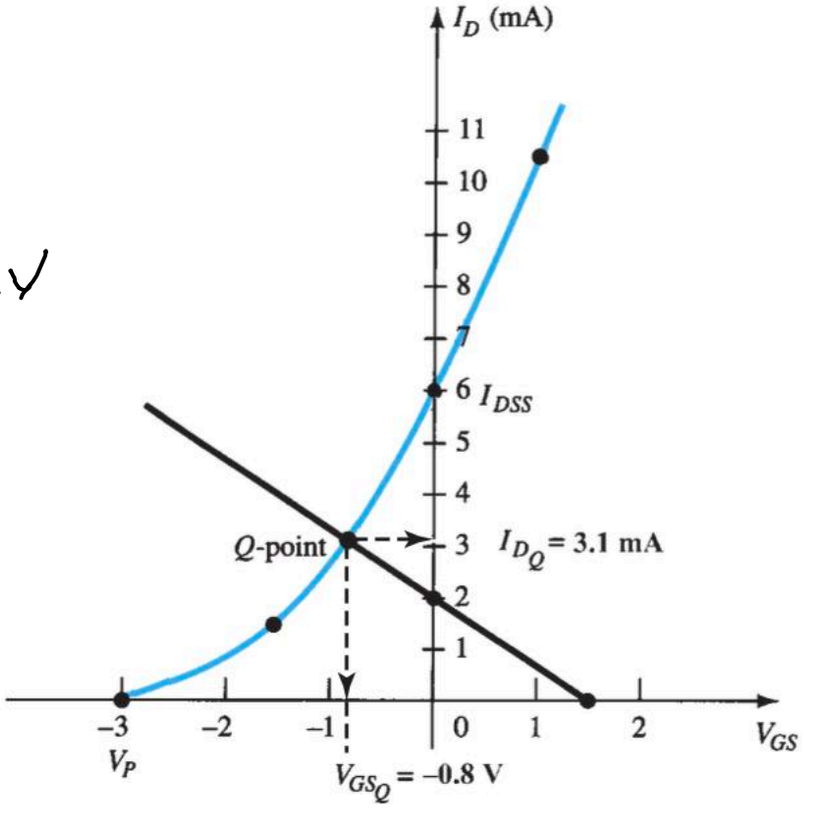


FIG. 7.31
Determining the Q-point for the network of Fig. 7.30.

For $V_{GS} = \pm 1V$

$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 = 10,67\,mA$

From KVL

$V_G = \frac{10\,M\Omega \cdot 18V}{10\,M\Omega + 110\,M\Omega} = 1,5V$

$V_{GS} = V_G - I_D \cdot R_S = 1,5V - I_D \cdot 750$

If $V_{GS} = 0 \Rightarrow I_D = 2\,mA$
If $I_D = 0 \Rightarrow V_{GS} = -1,5V$
} draw line

$V_{GSQ} = -0,8V \quad I_{DQ} = 3,1\,mA$

Finally $V_{DS} = V_{DD} - I_D(R_D + R_S) = 10,1V$

Mathematical Model Summary for Enhancement-type Mosfet

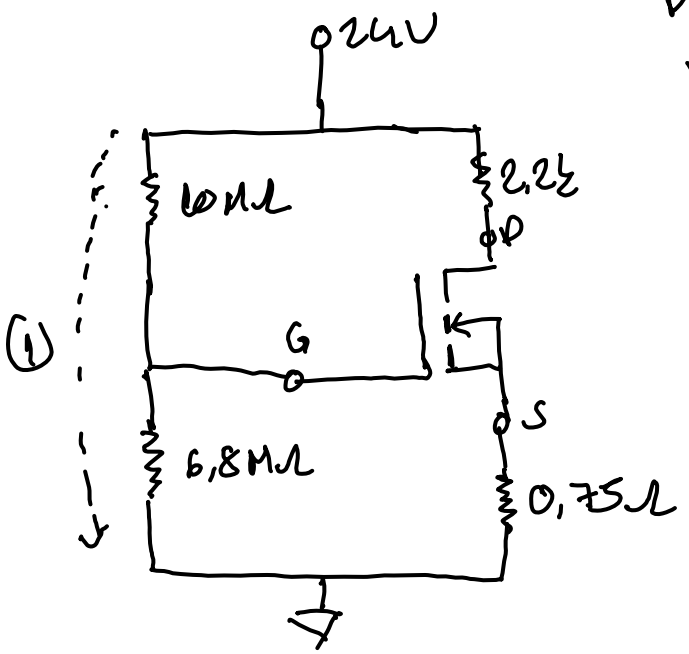
Cut-off Region: $I_{DS}=0$ for $V_{GS} \leq V_{GS(th)}$

Linear Region: $I_{DS}=k\left(V_{GS}-V_{GS(th)}-\frac{V_{DS}}{2}\right)V_{DS}$

for $V_{GS}-V_{GS(th)} \geq V_{DS} \geq 0$

Saturation Region: $I_{DS}=k\left(V_{GS}-V_{GS(th)}\right)^2$

for $V_{DS} \geq V_{GS}-V_{GS(th)}$



Determine I_{DQ}, V_{GSQ}, V_D and V_S

$V_{GS(th)} = 3V$, $I_{D(on)} = 5mA$, $V_{GS(on)} = 6V$

$$k = \frac{I_{D(on)}}{V_{GS(on)} - V_{GS(th)}} = \frac{5mA}{(6-3)^2} = \frac{5}{9} \cdot 10^{-3}$$

$$I_D = \frac{5}{9} \cdot 10^{-3} (V_{GS} - 3)^2 \Rightarrow 1^{st} \text{ curve}$$

① KVL: $24V \cdot \frac{6,8M\Omega}{10M\Omega + 6,8M\Omega} = 9,714 = V_G$

$V_{GS} = V_G - 0,75k\Omega \cdot I_D = 9,714 - 0,75k\Omega \cdot I_D$

$\Rightarrow 2^{nd}$ curve

check: $V_{DS} = 24V - (2,2k + 0,75k) I_D$

$= 9,25V$

$V_{DS} > V_{GS} - V_{GS(th)}$

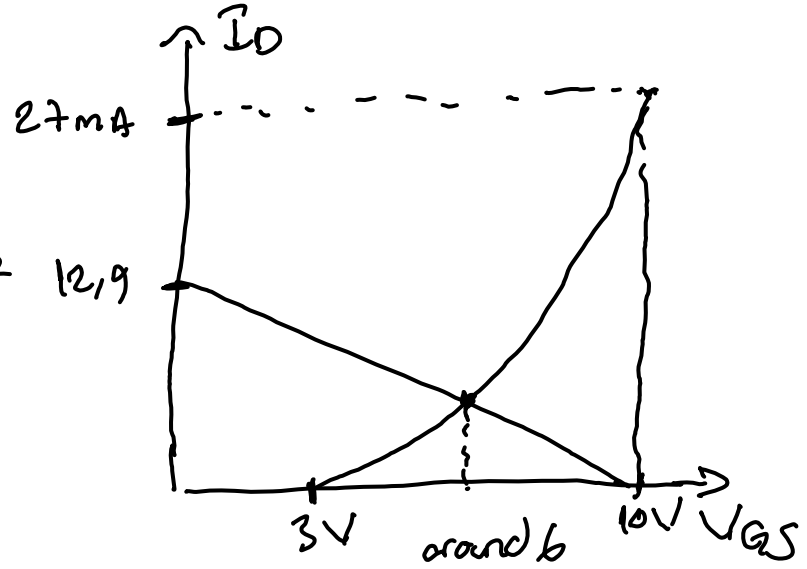
$9,25 > 3$ ✓

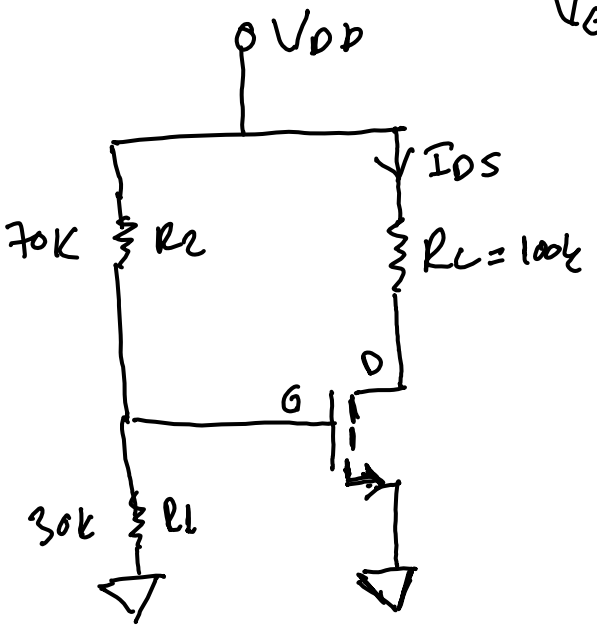
SAT

$V_{GSQ} = 6V$

$I_{DQ} = \frac{5}{9} \cdot 10^{-3} \cdot (6-3)^2$

$= 5mA$





$V_{GS(th)} = 1 \quad k = 1{,}25\ \mu A/V^2 \quad I_{DS} = ? \quad V_{DS} = ?$

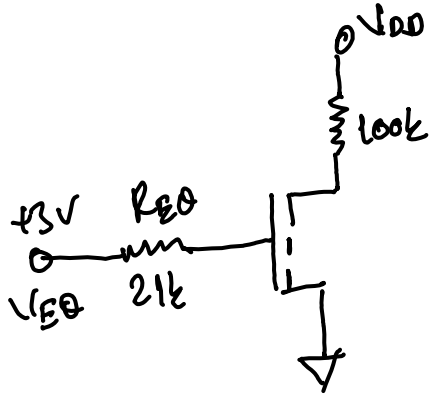
$V_{E\Theta} = \frac{R_1}{R_1+R_2} V_{DD} = 3V \qquad R_{E\Theta} = \frac{R_1 \cdot R_2}{R_1+R_2} = 21k$

① $V_{E\Theta} = I_G R_{E\Theta} + V_{GS} \Rightarrow I_G = 0, \ V_{GS} = 3V$

② $V_{DD} = I_{DS} \cdot R_L + V_{DS}$

Assume SAT

$I_{DS} = k(V_{GS} - V_{GS(th)})^2 = 12{,}5 \cdot 10^{-6} \cdot (3-1)^2 = 50\mu A$



② $V_{DS} = 10 - 50 \cdot 10^{-6} \cdot 10^5 = 5V$

$V_{GS} - V_{GS(th)} = 2V$

$V_{DS} > V_{GS} - V_{GS(th)}$ SAT ✓

Θ point is $I_{DS} = 50\mu A$

$V_{DS} = 5V$ with $V_{GS} = 3V$

This could have been solved graphically too!

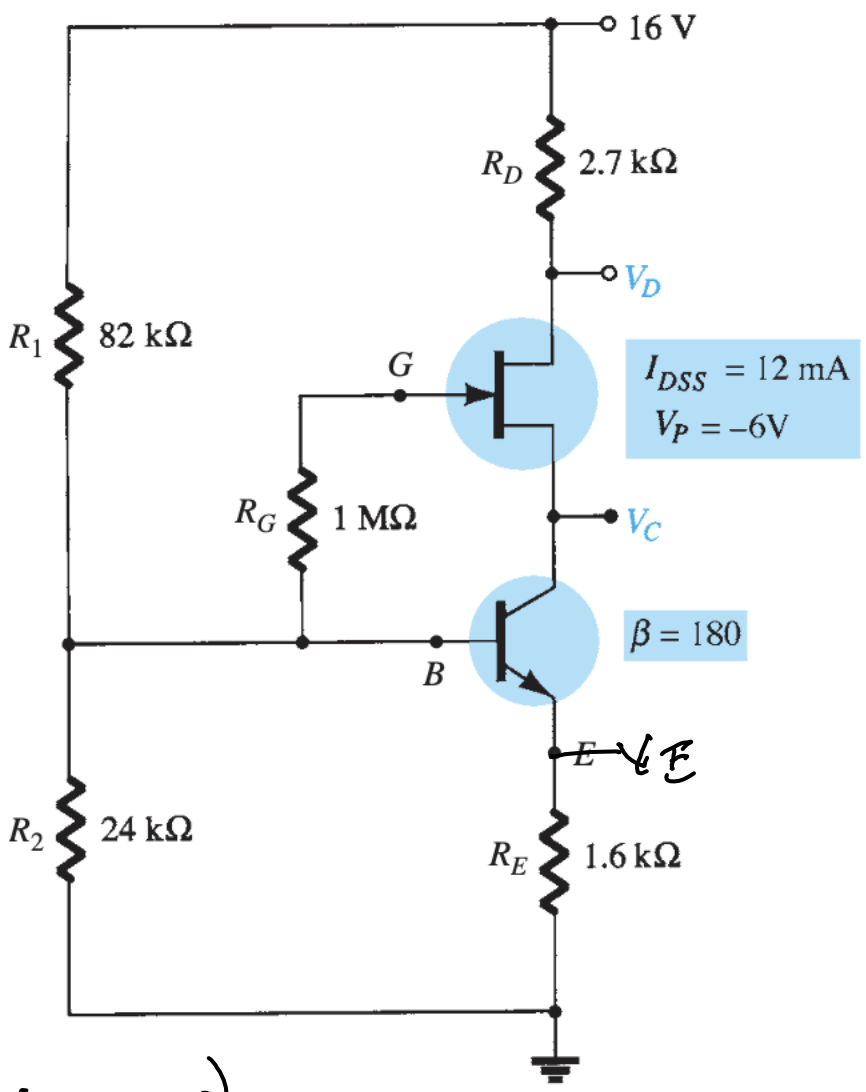


FIG. 7.46
Example 7.12.

V_D and V_C ?

(1) $V_B = \frac{16 \cdot 24k}{82+24} = 3,62V$ //

$V_E = 3,62 - 0,7 = 2,92V$

$I_E = \frac{2.92}{1.6k} = 1,825\,mA$

we can say that

$I_C \sim I_E$ & $I_D = I_S = I_C$

The voltage-divider configuration where approximate technique can be applied ($\beta R_E = 180 \times 1.6k\Omega = 288k\Omega > 10R_2 = 240k\Omega$)

(2) $V_D = 16 - 2,7k \cdot I_D \simeq 11,07V$

(3) $I_{DQ} = I_{DSS}\left(1 - \frac{V_{GSQ}}{V_P}\right)^2$ { $V_{GSQ} = -3,66V$

I_{DQ}: 1,825mA I_{DSS}: 12mA V_P: −6V

Knowing $I_G = 0$, $V_G = V_B$

$V_C = V_B - V_{GSQ} = 3,62 - (-3,66) = 7,32V$